# Dhruv Aggarwal

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## Profile

Experimental neuroscientist with an MSc (Distinction) in Brain Sciences from the University of Glasgow, specializing in cognitive neuroscience and empirical quantitative research. Experienced in programming and executing controlled multifactorial cognitive experiments (PsychoPy) and eye-tracking paradigms (iMotions). Proficient in advanced regression and Bayesian analyses (R) and computation using Python and MATLAB. My research interests include mapping the neural bases for cognitive functions such as memory, motivation, attention, and inhibitory control in humans.

## Education

**MSc Brain Sciences** | University of Glasgow, United Kingdom 09/2024 - 08/2025

- Grade: Distinction
- Scholarship: India Merit Scholarship worth £5000
- Thesis: "Associative Asymmetry in Episodic Memory: An Experimental Study on Directional Biases in Recall Using Picture Sequences"
    - Summary: Preregistered; Programmed in PsychoPy; behavioural measures of accuracy; 120 participants; Generalized Linear Mixed Modelling and Bayesian analysis in R

**BA (Hons.) Psychology** | O. P. Jindal Global University, India 08/2021 - 08/2024

- Grade: CGPA 8.0/8.0 | Gold Medal - 1st position in cohort
- Scholarship: Savitri Jindal Merit-cum-Means Scholarship (2021-2024)
- Thesis: "Internet Browsing Behaviour and its Influence on Cognitive Processing in College Students: A Study Among Indian Young Adults"
    - Summary: Programmed in PsyToolKit; behavioural and eye-tracking measures of cognitive control; 48 participants; Bayesian analysis in R

## Methodological & Technical Skills

- **Statistical analysis:** R (proficient), SPSS (proficient), Python (intermediate), MATLAB (intermediate)
- **Statistical techniques:** Bayesian and Advanced Regression Modelling in R (brms)
- **Experimental software:** PsychoPy (experienced), iMotions Eye-tracking Software (experienced)
- **Reproducible & Open Science:** PRISMA-ScR systematic review methodology, OSF pre-registration, Preprints
- **Data Collection & Management**: REDCap (proficient), Pavlovia (proficient)
- **Languages:** English (C2/Proficient - IELTS 8.5), Hindi (Mother tongue)

## Academic Publications, Manuscripts & Projects

***Peer-reviewed Publications:***

**Aggarwal, D**., Zhang, K., & Bartlett, J. (2026). Debunking decreases mistaken causal inferences: A direct replication of Irving et al. (2022). *Replication Research*, 2. https://doi.org/10.17879/replicationresearch-2026-10085

**Aggarwal, D.**, Naik, J., & Lindquist, D. H. (2025). Biphasic model of addiction: Neurobehavioral adaptations. *Current Behavioral Neuroscience Reports*, 12(1). https://doi.org/10.1007/s40473-025-00314-0

***Under Peer-review:***

**Aggarwal, D.**, & Smith, P. Intentional Forgetting of Autobiographical Memories: Current Neuroimaging Evidence and Moving Towards an Integrationist EEG-fMRI Approach. Preprint available at: https://doi.org/10.31234/osf.io/45brz_v3

**Aggarwal, D.**, & Lindquist, D. H. Internet Browsing Behaviour and Its Influence on Cognitive Processing in College Students. Preprint available at: https://doi.org/10.31234/osf.io/9qnaj_v1
**Aggarwal, D.**, Gupta, A., Tyagi, K., & Bhatia, D. Personal memory in the digital age: A scoping review of social media posting effects.

***Pre-registration:***
**Aggarwal, D**., & Bhardwaj, S. (2026, October 30). Verification Report: Mutual interplay between cognitive offloading and secondary task performance. Available at OSF: https://doi.org/10.17605/OSF.IO/W5BEA
Koukouvinis, S., Wimber, M., & **Aggarwal, D.** (2025, June 6). Directionality of associative memory for image sequences. Available at OSF: https://doi.org/10.17605/OSF.IO/TBGVC

## Research Experience

**Research Intern** | O. P. Jindal Global University, India | Remote 05/2026 - Present

- Supervisor: Dr. Kanika Malik | Funding: Wellcome Trust & King's College London
- Project: 'METROPOLIS (Mechanistic Trial of Problem Solving and Behavioural Activation for Depression)'
- Engaging in REDCap programming along with data wrangling, analysis, and visualization in R from the testing phase of the project.

**Research Assistant** | Emotion and Cognition Research Centre, JSPC, India | Hybrid 06/2024 - 06/2026

- Supervisor: Dr. Divya Bhatia | Funding: O. P. Jindal Global University
- Project: 'Personal Memory in the Digital Age: Scoping Review of Social Media Posting Effects'
- Led systematic scoping review following PRISMA-ScR guidelines; developed Boolean search strings, led dual-stage screening and data extraction across Scopus and PubMed.

## Teaching, Mentorship & Open Science

**Director of Research Initiatives** | Tertiary Education Advisors (TEA) | Remote 01/2026 - Present

- Co-developed the Trishul Research Program. Periodically auditing its governing academic policy to ensure compliance with international research standards and open science practices.
- Mentoring students in research design, academic writing, and open science within psychology and neuroscience.

**Reviewer for Open Science Curriculum** | Framework for Open and Reproducible Research Training (FORRT) | Remote 09/2025 - Present

- Contributing to international initiatives promoting transparent and reproducible research practices.
- Reviewing educational materials promoting open science, pre-registration, and reproducibility in research.

## Conferences & Presentations

**Oral Presentation:** "Associative Asymmetry in Episodic Memory: An Experimental Study on Directional Biases in Recall Using Picture Sequences" | **1st Prize** 09/2025
*3rd Annual Student Research Conference, Jindal School of Psychology, Sonipat, India*

**Poster Presentation**: "The Two-Stage Neuropsychological Model of Addiction: A Review" 12/2023
*Brain and Behaviour Conference, Bengaluru, India*

**Poster Presentation:** "Social Media, Affective Experiences, and Mental Health of Young Adults in Northern India" | **2nd Prize** 09/2023
*1st Annual Student Research Conference, Jindal School of Psychology, Sonipat, India*